

MPLADS AI INSIGHT — MASTER SOFTWARE SPECIFICATION

The 6 Mandatory Architecture & Product Documents Required Before Building with AI (PRD, TRD, App Flow, UI/UX Brief, Backend Schema, Implementation Plan)

Project Title:	MPLADS AI Insight Platform	Theme / Scope:	Smart Governance & Public Financial Oversight
Problem Statement:	MoSPI SIH 2026 — PS 26102	Document Type:	6-Document Architecture Master Spec
Lead Architect:	Kunal Bhalerao & Team Sentinel	Version / Date:	v1.0.0 Production Blueprint • 2026-08-28

DOCUMENT 1: PRODUCT REQUIREMENTS DOCUMENT (PRD)

One-Line Description: An AI-driven financial vigilance, anomaly detection, and decision-support platform designed to monitor, evaluate, and safeguard ₹8,300+ Cr annual MPLADS development expenditures across India.

Problem Statement: Under the MPLAD Scheme, MPs recommend ₹5 Cr per year of constituency works. Traditional oversight lacks cross-scheme reconciliation (overlap with PMGSY/JJM), photographic authenticity verification, and early warning for stalled or front-loaded expenditures.

Target Users: Ministry of Statistics & PI (National Officers), District Magistrates/Collectors (Nodal Authorities), Members of Parliament, and Citizens.

Core Value Proposition: Prevents double-funding across central schemes, identifies inflated vendor concentrations (HHI > 2500), extracts EXIF GPS geotags, and generates actionable on-site inspection dockets with zero black-box obscurity.

PRD Feature Priority Matrix (P0 = Critical, P1 = Important, P2 = Nice to Have)

Feature	Priority	User Role	Functional Description	Acceptance Criteria
Cross-Scheme Overlap Detection	P0	Collector / Officer	Scans geospatial proximity (<100m) to PMGSY, JJM, AMRUT, and SCM assets.	Flags asset within 50m with estimated duplicate financial exposure.
Physical-Financial Gap Analysis	P0	Officer / Auditor	Detects front-loaded voucher payments divergence from verified physical stage-gate.	Alert triggered if expenditure > 80% while physical progress < 30%.
Client-Side EXIF & pHash Audit	P0	Collector / Officer	Extracts hardware GPS coordinates from site photos and computes 64-bit dHash.	Flags distance > 100m from site and image similarity > 85%.
Allowlisted NL Search	P0	All Users	Natural language intent parser with allowlisted parameterized execution.	Translates NL queries to SQL parameters without raw SQL execution.
Inspection Docket Generator	P1	District Collector	Generates official jsPDF physical inspection dossiers with signature blocks.	Downloads formatted PDF with coordinates, checklist, and audit fields.
Citizen QR Grievance Loop	P1	Citizen / Public	Public transparency portal for tracking local works and filing GPS geotagged issues.	Submits feedback without exposing internal AI weights or raw algorithms.
Time-Series Fund Forecaster	P2	Ministry Officer	ARIMA/Prophet projection of annual fund utilization velocity.	Disabled with explicit notice if historical periods < 12 cycles.

DOCUMENT 2: TECHNICAL REQUIREMENTS DOCUMENT (TRD)

Architecture: 3-Tier Enterprise Architecture (React 19 + FastAPI + PostgreSQL 15/PostGIS).

Frontend Stack: React 19.2, TypeScript 6.0, Vite 8.2, Tailwind CSS v4, Lucide Icons, Recharts 3.10, React-Leaflet 5.0, jsPDF & autoTable, exifr.

Backend Stack: Python 3.11+, FastAPI, Pydantic v2, Uvicorn ASGI, SQLAlchemy 2.0, Scikit-Learn (RandomForest & IsolationForest), Pytest.

Database Stack: PostgreSQL 15 with PostGIS spatial extension; 17 tables with indexes on all foreign keys and partition fields.

Security: Strict Parameterized Query Execution (Zero Raw Dynamic SQL), Role-Based Access Control (RBAC), CORS Middleware, and Environment-Variable Secret Management.

DOCUMENT 3: APPLICATION FLOW & NAVIGATION MAP

User Navigation Pipeline:
[Landing Page] → [Role-Gated Login (/login)] → [Executive Dashboard] → [Project Explorer / Map View] → [Forensic Project Detail] → [AI Evidence Upload & pHash Audit] → [Inspection Docket PDF]

- **Ministry Officer Flow:** Full 15-page access (AI Analytics Lab, ML Pipeline, Sector Analytics, Data Sources Registry, Data Quality Monitor, Settings).
- **District Collector Flow:** Field vigilance queue, contractor HHI concentration radar, inspection docket dispatch, and local alert management.
- **Citizen Auditor Flow:** Constituency project finder, progress tracking, QR code verification simulator, and grievance filing.

DOCUMENT 4: UI/UX DESIGN BRIEF & SYSTEM DESIGN TOKENS

Design Philosophy: High-Trust Government Grade (Clean, uncluttered, accessible, high data density, zero startup gimmicks).
Color Palette Tokens: Primary MoSPI Blue (#003580), Gov Navy Dark (#002060), Saffron Accent (#FF6B00), Positive Green (#16a34a), Critical Risk Red (#dc2626), Warning Amber (#d97706), Surface Slate (#f1f3f6).
Typography: Segoe UI, -apple-system, BlinkMacSystemFont, Roboto, sans-serif (8pt to 20pt scale, strict 1.2-1.4 leading).
Component Hierarchy: App → TopBar (Language Toggle / Role Profile) → Sidebar (Role-Filtered Navigation) → PageHeader → KPI Cards Grid → Charts Row → Table → Evidence Modal.

DOCUMENT 5: BACKEND DATABASE SCHEMA & API SPECIFICATION

Database Tables (17): states, districts, constituencies, members, sectors, agencies, works, fund_releases, expenditures, work_progress, citizen_suggestions, district_indicators, risk_scores, anomaly_scores, priority_scores, data_sources, data_quality_errors, extraction_runs.
RESTful API Endpoints (18): /api/states, /api/districts, /api/constituencies, /api/members, /api/sectors, /api/agencies, /api/works, /api/works/{id}, /api/funds, /api/expenditure, /api/dashboard/summary, /api/ai/risk/{id}, /api/ai/anomalies, /api/ai/priorities, /api/search.

DOCUMENT 6: 12-PHASE IMPLEMENTATION PLAN & TRACEABILITY MATRIX

Req ID	Requirement Summary	Target Screen	API Endpoint	Database Table	Implementation Task
REQ-01	Role-Based Auth & Session	/login	AuthContext	members / users	Phase 4: Gated Auth Setup
REQ-02	Works Explorer & Filter	/projects	GET /api/works	works	Phase 6: Paginated Grid
REQ-03	EXIF GPS & pHash Verification	/projects/:id	Client-side Engine	works.evidence_phash	Phase 7: AI Photo Audit
REQ-04	Leaflet GIS Map & Overlaps	/map	GET /api/districts	works, districts	Phase 6: Leaflet GIS Layer
REQ-05	Allowlisted NL Search	TopBar / Search	POST /api/search	works (Parameterized)	Phase 7: Intent Parser
REQ-06	5-Factor MP Performance	/mp-analytics	GET /api/members	members, works	Phase 6: Composite Scoring
REQ-07	Inspection Docket jsPDF	/ai-analytics	Client-side jsPDF	works, risk_scores	Phase 8: Export Generator
REQ-08	Data Governance Registry	/data-sources	GET /api/sources	data_sources	Phase 10: Provenance View
REQ-09	Data Quality & ETL Monitor	/data-quality	GET /api/quality	data_quality_errors	Phase 10: ETL Monitor View

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